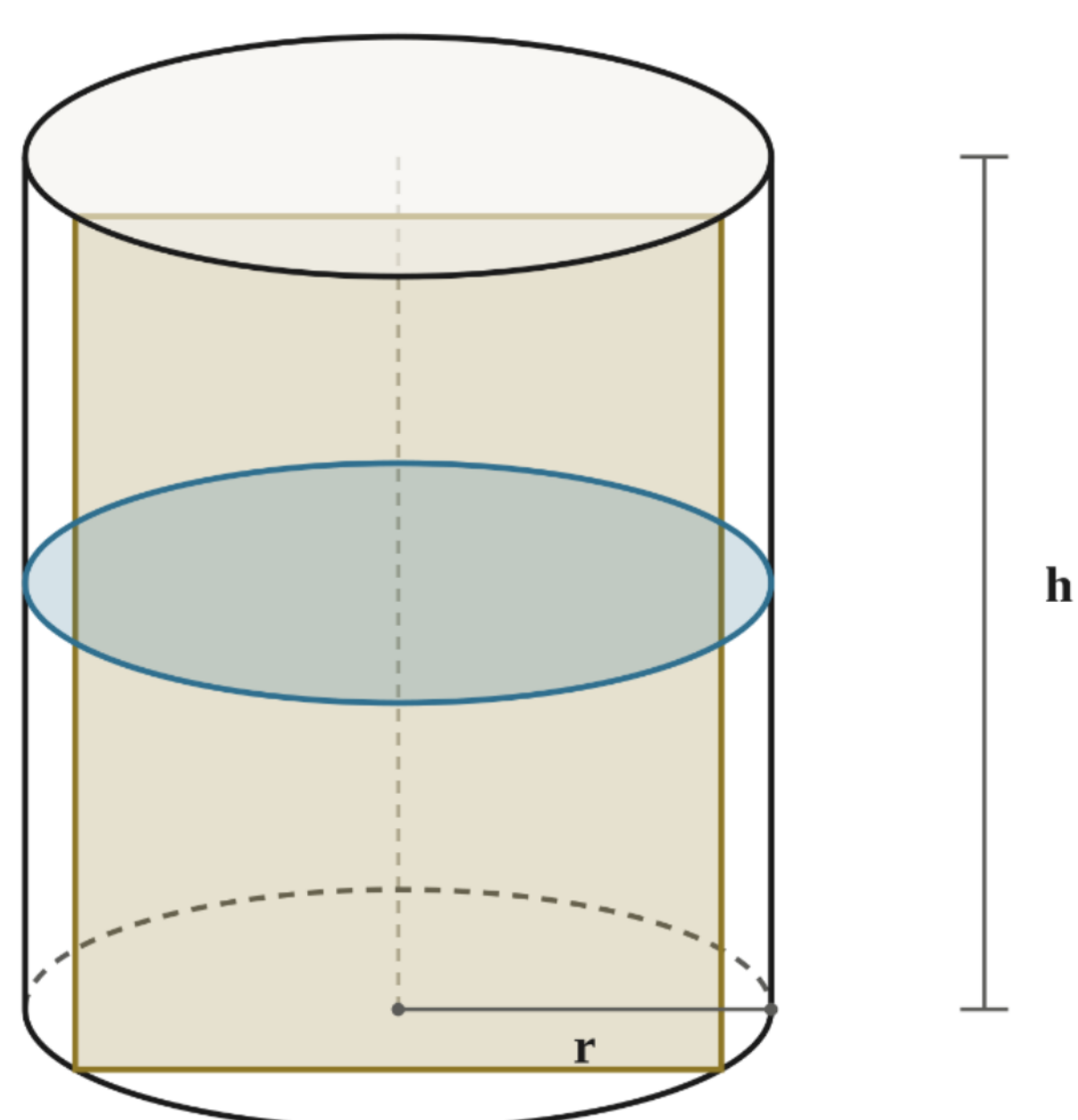
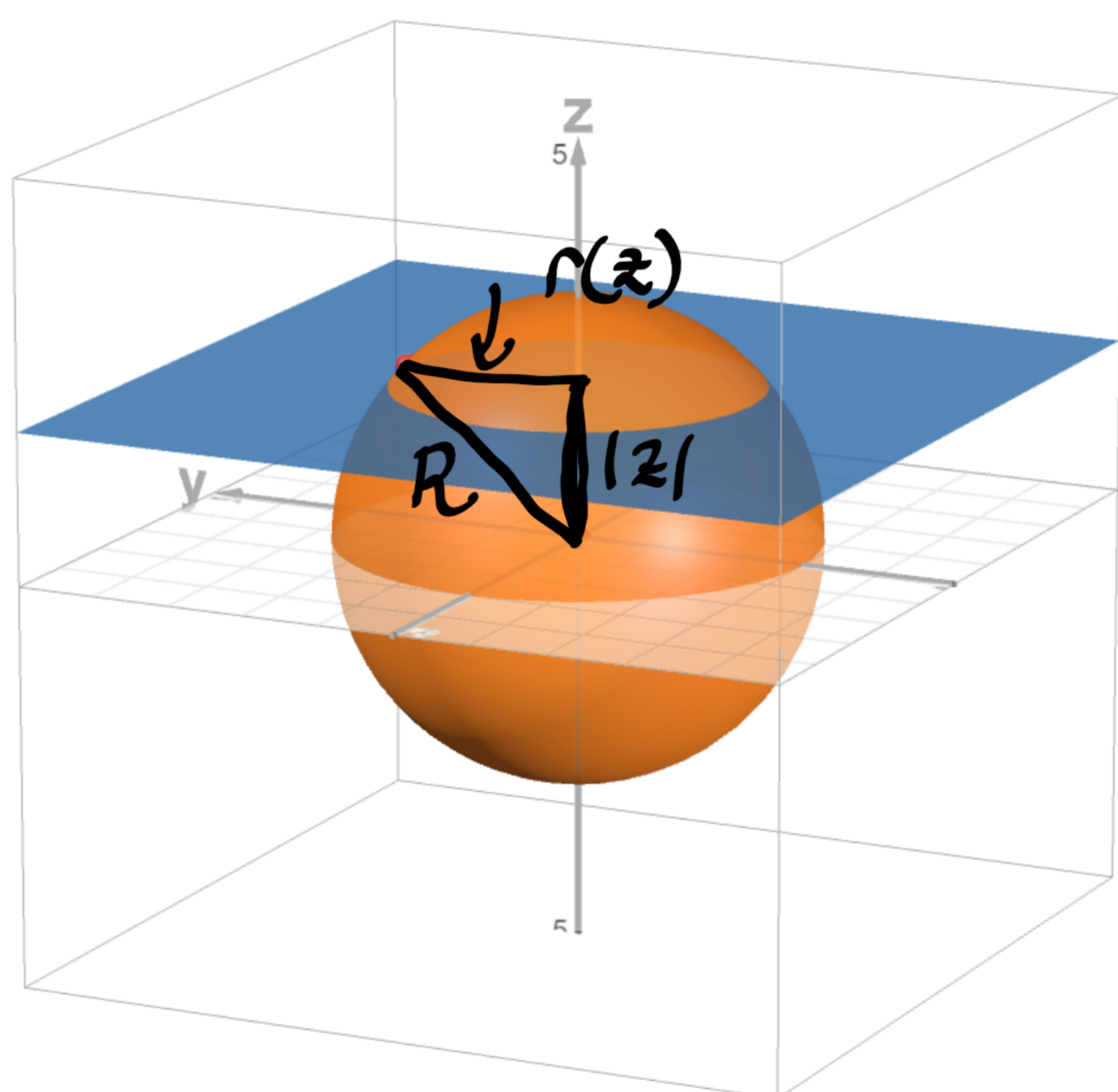


Right Circular Cylinder — Two Cross-Sections



■ Cross-section 1: _____
■ Cross-section 2: _____



$$\int_{-R}^R \underbrace{(\text{Area of a circle})}_{\pi(r(z))^2} dz$$

radius of circle

$$R^2 = (r(z))^2 + |z|^2$$

$$R^2 = (r(z))^2 + z^2$$

$$r(z) = \sqrt{R^2 - z^2}$$

$$\int_{-R}^R \pi \left(\sqrt{R^2 - z^2} \right)^2 dz = \int_{-R}^R \pi (R^2 - z^2) dz$$

$$= \frac{4}{3} \pi R^3$$